KAIZEN MATHS

Lesson starter

Algebraic substitution

Quadratic and cubic expressions

4 questions · about 8 minutes

Name _____

Substitute, then expand and simplify.

S1 Given $y = 3x - 2$

Write in terms of x and simplify

$2x^3 - 2x^2 + 5y + 1$

S2 Given $y = 5x - 2$

Write in terms of x and simplify

$6x^2 + 4y + 6$

S3 Given $y = 3x$

Write in terms of x and simplify

$3x^2 + 5xy + 5y - 4$

S4 Given $y = 3x$

Write in terms of x and simplify

$4x^2 + 3xy + 3y + 2$

Show your working

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AS-08 · S · Starter

Show your working

KAIZEN MATHS

Exit ticket

Algebraic substitution

Quadratic and cubic expressions

Name _____

Simplify each expression.

E1 Given $y = 3x$

Write in terms of x and simplify

$$3x^2 + 5xy + 5y - 4$$

E2 Given $y = 3x$

Write in terms of x and simplify

$$4x^2 + 3xy + 3y + 2$$

AS-08 · E · 2 questions

Show your working

KAIZEN MATHS

Exit ticket

Algebraic substitution

Quadratic and cubic expressions

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AS-08 · E · 2 questions

Show your working

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AS-08 · E · 2 questions

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Write in terms of x and simplify

$3x^2 + 5xy + 5y - 4$

E2 Given $y = 3x$

Write in terms of x and simplify

$4x^2 + 3xy + 3y + 2$

AS-08 · E · 2 questions

Show your working

Teacher answers & working

AS-08 · Quadratic and cubic expressions

Page 1: two starters. Page 2: four exit tickets. Cut along the dashed lines.

Print student pages single-sided at 100%. Match S and E labels to the solutions below.

S1

Given y=3x-2

$$2x^3 - 2x^2 + 5y + 1$$

Write in terms of x and simplify

Substitute y=3x-2.

$$2x^3 - 2x^2 + 5(3x - 2) + 1$$

$$2x^3 - 2x^2 + 15x - 10 + 1$$

$$2x^3 - 2x^2 + 15x - 9$$

S2

Given y=5x-2

$$6x^2 + 4y + 6$$

Write in terms of x and simplify

Substitute y=5x-2.

$$6x^2 + 4(5x - 2) + 6$$

$$6x^2 + 20x - 8 + 6$$

$$6x^2 + 20x - 2$$

S3 / E1

Given y=3x

$$3x^2 + 5xy + 5y - 4$$

Write in terms of x and simplify

Substitute y=3x.

$$3x^2 + 5x(3x) + 5(3x) - 4$$

$$3x^2 + 15x^2 + 15x - 4$$

$$(3 + 15)x^2 + 15x - 4$$

$$18x^2 + 15x - 4$$

S4 / E2

Given y=3x

$$4x^2 + 3xy + 3y + 2$$

Write in terms of x and simplify

Substitute y=3x.

$$4x^2 + 3x(3x) + 3(3x) + 2$$

$$4x^2 + 9x^2 + 9x + 2$$

$$(4 + 9)x^2 + 9x + 2$$

$$13x^2 + 9x + 2$$